

Decent Coaching Classes

IX CBSE

Mind Your Math - VII

Marks: 80

Time: 3 Hrs

General Instructions: (1) All questions are compulsory. (2) The question paper has five sections: A, B, C, D and E. (3) Section A has 20 MCQs of 1 mark each; Section B has 5 questions of 2 marks each; Section C has 6 questions of 3 marks each; Section D has 4 questions of 5 marks each; Section E has 3 case-based questions of 4 marks each. (4) Show all steps. Use $\pi = 22/7$ where required.

SECTION – A

20 × 1 = 20 marks

1. The point $(-3, 4)$ lies in the _____ quadrant.
(A) I (B) II (C) III (D) IV
2. The coordinates of the point on the y-axis, 5 units below the origin, are _____.
(A) $(5, 0)$ (B) $(0, 5)$ (C) $(-5, 0)$ (D) $(0, -5)$
3. The zero of the linear polynomial $3x - 12$ is _____.
(A) -4 (B) 4 (C) 12 (D) 3
4. The graph of $y = 2x + 1$ cuts the y-axis at the point _____.
(A) $(1, 0)$ (B) $(0, 2)$ (C) $(0, 1)$ (D) $(-1/2, 0)$
5. Which of the following is an irrational number?
(A) $\sqrt{16}$ (B) $22/7$ (C) $0.3333...$ (D) $\sqrt{7}$
6. The decimal expansion of $7/8$ is _____.
(A) terminating (B) non-terminating repeating
(C) non-terminating non-repeating (D) none of these
7. The value of $\sqrt{2} \times \sqrt{8}$ is _____.
(A) 2 (B) 4 (C) $\sqrt{10}$ (D) 16
8. The value of 103×97 is _____.
(A) 10009 (B) 9999 (C) 9991 (D) 9901
9. If $a + b = 7$ and $ab = 12$, then $a^2 + b^2 =$ _____.
(A) 19 (B) 25 (C) 37 (D) 49
10. If $x + 1/x = 3$, then $x^2 + 1/x^2 =$ _____.
(A) 5 (B) 9 (C) 7 (D) 11
11. Chords of a circle which are equal in length are _____ from the centre.
(A) at unequal distances (B) at equal distances
(C) at double distances (D) at half distances
12. An arc of a circle subtends an angle of 100° at the centre. The angle it subtends at a point on the remaining part of the circle is _____.
(A) 50° (B) 100° (C) 200° (D) 25°
13. A chord of length 8 cm is drawn in a circle of radius 5 cm. Its distance from the centre is _____.
(A) 2 cm (B) 3 cm (C) 4 cm (D) 6 cm
14. The area of a triangle with sides 3 cm, 4 cm and 5 cm is _____.
(A) 6 cm^2 (B) 12 cm^2 (C) 7.5 cm^2 (D) 10 cm^2
15. The circumference of a circle is 44 cm. Its radius is _____. ($\pi = 22/7$)

- (A) 14 cm (B) 7 cm (C) 22 cm (D) 3.5 cm
16. The area of an equilateral triangle of side 4 cm is _____.
 (A) $8\sqrt{3} \text{ cm}^2$ (B) 16 cm^2 (C) $4\sqrt{3} \text{ cm}^2$ (D) $2\sqrt{3} \text{ cm}^2$
17. The probability of a sure event is _____.
 (A) 0 (B) 1 (C) $1/2$ (D) -1
18. A die is thrown once. The probability of getting a number greater than 4 is _____.
 (A) $1/6$ (B) $1/2$ (C) $2/3$ (D) $1/3$
19. The next term of the sequence 3, 7, 11, 15, ... is _____.
 (A) 17 (B) 18 (C) 19 (D) 20
20. The sum of the first 10 natural numbers is _____.
 (A) 45 (B) 55 (C) 50 (D) 60

SECTION – B

$5 \times 2 = 10$ marks

21. The points A(2, 3) and B(2, -5) are joined by a line segment. Find the length of AB and state whether AB is parallel to the x-axis or the y-axis. [2]
22. Express 0.363636... (the digits 36 repeat forever) in the form p/q , where p and q are integers and $q \neq 0$. [2]
23. Factorise: $8a^3 + 27b^3$ [2]
24. A bag contains 3 red balls and 5 blue balls. One ball is drawn at random. Find the probability that the ball drawn is (i) red (ii) not red. [2]
25. Find the 15th term of the A.P. 7, 10, 13, 16, ... [2]

SECTION – C

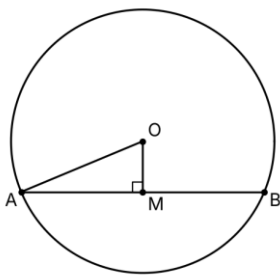
$6 \times 3 = 18$ marks

26. If $x = 2$ is a zero of the linear polynomial $p(x) = kx - 6$, find the value of k. Hence find $p(5)$. [3]
27. Rationalise the denominator of $1/(\sqrt{5} - \sqrt{2})$ and simplify. [3]
28. Expand: $(2x - 3y + z)^2$ [3]
29. A chord AB of length 16 cm is drawn in a circle of radius 10 cm. Find the distance of the chord from the centre of the circle. [3]
30. Find the area of a triangle whose sides are 13 cm, 14 cm and 15 cm, using Heron's formula. [3]
31. Find the sum of the first 20 terms of the A.P. 3, 7, 11, 15, ... [3]

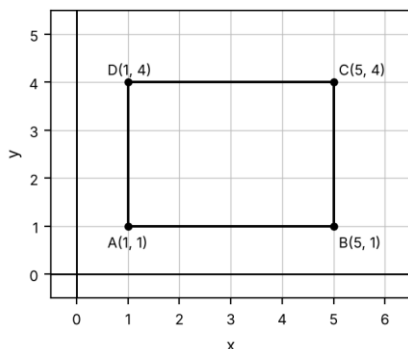
SECTION – D

$4 \times 5 = 20$ marks

32. Solve the following.
 (a) Factorise: $2x^2 - 5x - 3$ [2]
 (b) If $a + b + c = 9$ and $ab + bc + ca = 26$, find the value of $a^2 + b^2 + c^2$. [3]
33. Prove that the perpendicular drawn from the centre of a circle to a chord bisects the chord. [3]
 In the figure, O is the centre of the circle of radius 13 cm and $OM \perp AB$. If $AB = 24$ cm, find OM. [2]



34. In quadrilateral ABCD, $AB = 3$ cm, $BC = 4$ cm, $CD = 5$ cm, $DA = 6$ cm and $\angle B = 90^\circ$. Find (i) the perimeter of ABCD (ii) the length of the diagonal AC (iii) the area of the quadrilateral ABCD. [5]
35. The vertices of a rectangle ABCD are $A(1, 1)$, $B(5, 1)$, $C(5, 4)$ and $D(1, 4)$, as shown on the graph.



- (i) Write the lengths of AB and BC. [2]
- (ii) Find the area and the perimeter of the rectangle. [2]
- (iii) Write the coordinates of the point where the diagonals AC and BD meet. [1]

SECTION – E

$3 \times 4 = 12$ marks (Case-based questions)

36. Case Study: A class teacher surveyed 40 students of Class IX about their favourite sport. The results are shown below.
- Cricket: 16 students Football: 10 students Kabaddi: 8 students Others: 6 students
- One student is chosen at random from the class. Find the probability that the student
- (i) likes cricket. [1]
- (ii) likes football or kabaddi. [1]
- (iii) does not like football. [1]
- (iv) likes neither cricket nor football. [1]
37. Case Study: Ravi saves ₹50 in the first week. He decides to increase his saving by ₹10 every week, i.e. ₹50, ₹60, ₹70, ...
- (i) Show that his weekly savings form an A.P. and write the common difference. [1]
- (ii) How much does he save in the 8th week? [1]
- (iii) Find his total saving in 12 weeks. [1]
- (iv) In which week will he save ₹200? [1]
38. Case Study: A circular park has radius 14 m. A path 3.5 m wide runs all around it on the outside. The path is to be paved at ₹40 per m^2 . ($\pi = 22/7$)
- (i) Find the circumference of the park. [1]
- (ii) Find the area of the park. [1]
- (iii) Find the area of the path. [1]
- (iv) Find the total cost of paving the path. [1]